

A Faithful- C_9 Tensor Exclusion through Base Order 188 for Navarro's Character-Field Question

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Abstract

For a finite group A , an irreducible character α of A , and a faithful linear character λ of C_9 , consider the external product $\alpha \boxtimes \lambda$ of $A \times C_9$. An exact exhaustive GAP calculation shows that when $|A| \leq 188$, every such character of odd degree or odd character-field conductor has a single value that generates its full character field. The calculation covers all 4,423 base-group isomorphism types in the Small Groups Library, all 178,870 irreducible base characters, and all 1,073,220 choices after the six faithful C_9 factors are included. Thus the faithful- C_9 external-product mechanism producing the known order-1701 counterexample cannot produce a smaller counterexample. This is only a finite exclusion for that character template: the minimum order of a group failing the property, and in particular whether it equals 1701, remain unresolved.

1 The restricted question

For a complex character χ of a finite group G , write

$$\mathbb{Q}(\chi) = \mathbb{Q}(\{\chi(g) : g \in G\}).$$

Its conductor is the least positive integer m for which $\mathbb{Q}(\chi) \subseteq \mathbb{Q}(\zeta_m)$. Following Thiel [1], say that χ has the single-value property if some $g \in G$ satisfies

$$\mathbb{Q}(\chi(g)) = \mathbb{Q}(\chi).$$

The property FS₂ asks for this equality for every irreducible character having odd degree or odd conductor (the latter condition is also called 2-rationality).

Thiel gave explicit nonsolvable and solvable counterexamples to this question [1]. The solvable example is

$$\text{SmallGroup}(189, 5) \times C_9,$$

of order 1701, with a degree-3 external-product character. That work did not pursue a systematic minimum-order search and explicitly left open the possibility of a smaller counterexample. If

$$\mu = \min\{|G| : G \text{ does not satisfy FS}_2\},$$

then the example gives $\mu \leq 1701$. The result below does not supply the reverse inequality. Instead, it exhausts exactly the direct-product template adjacent to this known example.

Theorem 1 (Exact faithful- C_9 tensor exclusion). *Let A be any finite group with $|A| \leq 188$, let $\alpha \in \text{Irr}(A)$, and let $\lambda \in \text{Irr}(C_9)$ be any faithful linear character. Set $\chi = \alpha \boxtimes \lambda \in \text{Irr}(A \times C_9)$. If $\chi(1)$ is odd or the conductor of $\mathbb{Q}(\chi)$ is odd, then there is an $(a, c) \in A \times C_9$ such that*

$$\mathbb{Q}(\chi(a, c)) = \mathbb{Q}(\chi).$$

The tensor products in the theorem have orders from 9 through 1692. The theorem does not cover the three nonfaithful linear characters of C_9 , does not say that every group $A \times C_9$ in the range satisfies FS_2 , and says nothing about groups outside this character template.

2 Structural reductions

Let $C_9 = \langle c \rangle$, put $\zeta = \zeta_9$, and write $\lambda(c) = \zeta^k$, where

$$k \in \{1, 2, 4, 5, 7, 8\}.$$

Lemma 2 (Field and faithful-factor reduction). *If $d = \alpha(1)$ and $\chi = \alpha \boxtimes \lambda$, then*

$$\mathbb{Q}(\chi) = \mathbb{Q}(\{\alpha(a) : a \in A\}, \zeta).$$

Moreover, as k ranges over the six faithful exponents, the set of values of χ is unchanged.

Proof. For $a \in A$ and $0 \leq j < 9$,

$$\chi(a, c^j) = \alpha(a)\zeta^{kj}.$$

Taking $j = 0$ recovers every value of α . Taking $a = 1$ gives $d\zeta^{kj}$, and division by the nonzero rational integer d recovers ζ , because k is a unit modulo 9. This proves the field identity. Multiplication by any unit k permutes all residues modulo 9, which proves the assertion about value sets. $\square$

Lemma 3 (Linear base characters). *If α is linear, then $\alpha \boxtimes \lambda$ has the single-value property.*

Proof. The tensor product is a linear character, hence its image is a finite cyclic subgroup of $\mathbb{C}^\times$. Choose a group element whose image generates that cyclic group. Its character value generates the field of all values. $\square$

These lemmas both reduce the computation and explain why testing one faithful C_9 exponent suffices while the final accounting includes all six.

3 Exact exhaustive certificate

The computation was performed in GAP 4.15.1 [2], using SmallGrp 1.5.4 [3]. For every $1 \leq n \leq 188$, the program calls `NumberSmallGroups(n)` and then processes `SmallGroup(n,i)` at every catalogue index. The documented library coverage includes every order in this interval. For each representative A , the program constructs its character table and iterates over the full list $\text{Irr}(A)$.

For nonlinear α , the exact GAP field

$$F = \mathbb{Q}(\{\alpha(a) : a \in A\}, \zeta_9)$$

is constructed from cyclotomic values. The character is in the theorem's eligible class precisely when $\alpha(1)$ is odd or the conductor of F is odd. If $[F : \mathbb{Q}] = 6$, containment of $\mathbb{Q}(\zeta_9)$ forces $F = \mathbb{Q}(\zeta_9)$, and $\alpha(1)\zeta_9^k$ is already a generating value. In every other eligible case, the program exhausts the finite set

$$\{\alpha(a)\zeta_9^j : a \in A, 0 \leq j < 9\}.$$

The degree of a single cyclotomic value x is evaluated exactly. If m is its conductor, GAP applies the automorphisms $\sigma_r(\zeta_m) = \zeta_m^r$ for $r \in (\mathbb{Z}/m\mathbb{Z})^\times$, and the calculation uses

$$[\mathbb{Q}(x) : \mathbb{Q}] = \frac{\varphi(m)}{|\{r \in (\mathbb{Z}/m\mathbb{Z})^\times : \sigma_r(x) = x\}|}.$$

Every tested value lies in F , so equality of this exact degree with $[F : \mathbb{Q}]$ proves equality of the two fields. No floating-point arithmetic or numerical field recognition is involved.

The exhaustive accounting is shown in Table 1. The rows form a partition of all six faithful factors for all irreducible base characters.

Table 1: Exact partition of the faithful- C_9 tensor characters.

Case	Number of tensor characters
Linear, settled by Lemma 3	647,340
Nonlinear and ineligible	201,864
Eligible with full-field degree 6	154,080
Eligible with an exact generator search	69,936
Total	1,073,220
Failures among eligible characters	0

In total, the run covers 188 consecutive orders, 4,423 group isomorphism types, and 178,870 irreducible base characters. A separate log parser verifies consecutive order and catalogue-index coverage, all per-group and per-order accounting identities, the global totals, and the absence of a failure record. As an additional check of the field-degree routine, a second GAP verifier constructs each candidate field $\mathbb{Q}(\alpha(a)\zeta_9^j)$ directly and compares it with F ; it obtains the same partition and zero failures. These exact finite checks, together with Lemmas 2 and 3, prove Theorem 1.

For an adjacent example of proof-producing finite universal exclusion by exact computation in a different area, see [4]. That work is cited only for broad computational methodology and is not a source for the character-theoretic argument above.

4 Boundary control and limitations

The natural endpoint 188 is dictated by the reported example: $9 \cdot 188 = 1692 < 1701$, while its base group has order 189. The accompanying positive-control computation reconstructs

$$H = \text{SmallGroup}(189, 5) \times C_9$$

and the external product of GAP character 15 of the first factor with GAP character 2 of C_9 . It checks that the resulting irreducible character has degree 3 and

$$\mathbb{Q}(\chi) = \mathbb{Q}(\zeta_{63}), \quad [\mathbb{Q}(\chi) : \mathbb{Q}] = 36,$$

whereas the maximum degree of a field generated by one of its 261 values is 12. The exact single-value degree histogram is

$$1^{(75)}, \quad 2^{(12)}, \quad 4^{(12)}, \quad 6^{(18)}, \quad 12^{(144)}.$$

Thus the boundary instance fails, in agreement with Thiel’s construction [1], while every instance of the same faithful- C_9 template with base order at most 188 passes.

This comparison is a mechanism-specific finite exclusion, not a minimum-order theorem. A smaller counterexample could use a nonfaithful C_9 factor, a different direct-product factor, a character not of this external-product form, or a group with no relevant direct-product decomposition. Consequently the value of μ and the proposed equality $\mu = 1701$ remain unresolved. No claim of bibliographic priority is made for the finite exclusion.

Reproducibility and assistance disclosure

The decisive files in the accompanying source distribution are `evidence/scan_c9_bases_1e188.g` and its saved output, together with `evidence/audit_scan_c9_log.py`. From the project root, the main calculation and structural audit are run by

```
gap -q evidence/scan_c9_bases_1e188.g \  
  > evidence/scan_c9_bases_1e188.out 2>&1  
python3 evidence/audit_scan_c9_log.py
```

The order-1701 positive control is reproduced by

```
gap -q evidence/reproduce_1701.g \  
  > evidence/reproduce_1701.out 2>&1
```

The saved runs used GAP 4.15.1 on `aarch64-apple-darwin25-default64-kv10`, SmallGrp 1.5.4, and, for the positive control, CTblLib 1.3.11.

OpenAI language models assisted with the computational search design, the GAP and Python code, formulation and checking of the argument, and drafting of this manuscript. The reported certificate consists of exact computer algebra and exhaustive catalogue accounting; language-model output is not treated as mathematical evidence. A separate review pass reran the calculation and checked the central claim with a distinct direct-field comparison.

References

- [1] U. Thiel, On a question by G. Navarro on the character field generation of odd-degree characters, *arXiv preprint* arXiv:2608.02266v1 (2026), <https://arxiv.org/abs/2608.02266>.
- [2] The GAP Group, *GAP—Groups, Algorithms, and Programming*, Version 4.15.1, <https://www.gap-system.org>.
- [3] H. U. Besche, B. Eick, and E. A. O’Brien, *SmallGrp: The GAP Small Groups Library*, Version 1.5.4, <https://docs.gap-system.org/pkg/smallgrp/doc/manual.pdf>.
- [4] Z. Zeng, One-sided parity compression for certified SAT attacks on the 33-point Erdős–Szekeres problem, *SSRN working paper* 7334758 (2026), <https://doi.org/10.2139/ssrn.7334758>.